CSE 4630: Containerization for Developers

Department of Computer Science and Information Technology

7th Semester

Assignment 4

Book Reference	Chapters 10–11

Practical Questions

Q. No.	Question
1	A college maintains student records using MariaDB. The database administrator has prepared an SQL script containing the database schema and initial student records. Develop a Docker Compose application that automatically initializes the database when it is deployed for the first time. a. Create an initialization SQL file containing a students table and at least five records. b. Configure MariaDB to execute the SQL file automatically during initial deployment. c. Store database files in a named Docker volume. d. Modify the initialization SQL file and restart the container. Determine whether the modified SQL script executes again.
2	A PostgreSQL database contains important departmental records. The administrator wants to ensure that the database can be recovered even if its original Docker volume becomes unavailable. a. Deploy PostgreSQL using Docker Compose and create a database containing at least two related tables. b. Insert sample records and generate a logical backup using pg_dump. c. Delete the original database container and its data volume after safely preserving the backup. d. Deploy a new PostgreSQL container with a fresh volume and restore the database using the backup file.
3	A department uses MongoDB to store research-project information. Currently, every application connecting to the database has unrestricted administrative access, which is undesirable. a. Deploy a MongoDB container with authentication enabled. b. Create a database named research containing a collection of at least five projects. c. Create a database user with read/write access restricted to the research database. d. Demonstrate successful access using the authorized user.
4	A web application uses Redis to store frequently accessed information. To improve availability, the administrator deploys two Redis containers: a primary server and a replica. a. Create a Docker Compose application containing a Redis primary and replica. b. Configure the replica to obtain data from the primary using its Compose service name. c. Insert at least five key-value pairs into the primary and verify their availability on the replica. d. Attempt to modify a key directly on the replica and record the result.
5	A Node.js application contains several external dependencies. Developers frequently modify its source code, but rebuilding the Docker image repeatedly takes considerable time. a. Create a Node.js application using at least two external npm packages. b. Develop an initial Dockerfile that copies all application files before installing dependencies.

	c. Build the image and record the relevant build steps. d. Modify only the application source and rebuild the image. e. Redesign the Dockerfile to copy dependency manifests before installing npm packages. f. Rebuild after another source modification and compare the build-cache behavior.
6	A Java application runs successfully on the host system but experiences performance problems when executed inside a resource-constrained Docker container. a. Develop a Java application that performs a computationally intensive operation and reports its execution time. b. Package the application using an appropriate Java runtime image. c. Run the container without explicit resource restrictions and record its execution time. d. Execute the same container with CPU and memory restrictions. e. Inspect container resource utilization using docker stats.
7	A department has developed a PHP-based application. The administrator wants to separate the HTTP web server from the PHP execution environment rather than installing both components in one container. a. Create a Docker Compose application containing separate Nginx and PHP-FPM services. b. Develop a PHP page that accepts two numbers and displays their sum. c. Configure Nginx to forward PHP requests to the PHP-FPM service using its Compose service name. d. Ensure that only Nginx publishes a port to the host. e. Verify that the PHP application executes correctly through the browser. f. Stop the PHP-FPM container and observe the behavior of the application.
8	A research application was originally developed using a particular Python version and several third-party libraries. The application works on the developer's machine but fails on another computer because the installed library versions differ. a. Develop a Python application using at least two third-party libraries. b. Create a requirements.txt file specifying the required dependency versions. c. Build a Docker image using a specific Python runtime version. d. Execute the application inside the container and verify its output. e. Modify one dependency version and investigate whether the application behavior or dependency installation changes.